

Title. Cultural Common Ground and Four-Year-Olds' Referential Informativeness

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Cultural Common Ground and Four-Year-Olds' Referential Informativeness

From a young age, linguistic reference is a ubiquitous activity (Bruner, 1983). Appropriate reference requires that children use sufficiently informative referring forms (Grice, 1975). For example, a simple pointing gesture sometimes suffices for listeners to infer one's intended referent. But other times, a more informative form is required, like a modified noun phrase (NP). What influences children's choice to produce, say, *the red ball* instead of *it* or simply a pointing gesture?

One influential factor is COMMON GROUND. Common ground is propositionally structured background knowledge that interlocutors pragmatically infer and accept as true (Stalnaker, 2002). One way in which common ground differs from related notions of mutual knowledge in game theory (Lewis, 1969) and discourse reference in linguistics (Karttunen, 1968; also, Heim, 1983) is in its function of contextualizing the semantic interpretation of utterances (Stalnaker, 1978). This makes common ground a particularly useful tool for coordination in interdependent games (Gauthier, 1975), like reference (H. H. Clark & Wilkes-Gibbs, 1986), because it constrains semantic interpretation and can, itself, be modified on-the-fly (Lewis, 1979). Common ground has a recursive logical structure (Stalnaker, 2002), something which some philosophical (Schiffer, 1972) and psychological accounts (H. H. Clark & Marshall, 1981) have tried to skirt. Indeed, there remains discussion about the types of common ground (Gilbert, 2007) and its relation to other notions like mutual knowledge (Lee, 2001). In developmental psychology, common ground is often described broadly as shared knowledge, beliefs, and attitudes (Bohn & Köymen, 2018; E. V. Clark, 2015; Tomasello, 2019). This definition is sufficient for our purposes and is assumed throughout this article.

Basically all prior research has targeted PERSONAL COMMON GROUND (H. H. Clark, 1996; Tomasello, 2019), the content of this type of common ground is based in joint attention. Substantial

experimental research suggests that personal common ground influences linguistic reference in children as young as two years old (Vasil, 2023) through to adulthood (H. H. Clark, 1996). For example, two-year-olds in O'Neill (1996) either jointly attended with their parent to an experimenter hiding a toy in an opaque container, or else watched the hiding event solo. If they had watched the hiding event jointly, children often referenced the toy nonlinguistically, by pointing. In contrast, if they watched solo, children later used lexical nouns to refer to the toy. In short, there was an inverse relation between children's inferred, personal common ground and their linguistic referential informativeness, whereby "when the thoughts of... speakers are the same, the role of speech is reduced to a minimum" (Vygotsky, 1986, p. 238). Indeed, this inverse relation summarizes most prior research on children's common ground and referential informativeness (Vasil, 2023).

Despite its importance, personal common ground is just one type of common ground. Another type of common ground is INFERRABLE (Prince, 1981), which is "logical[ly] – or, more commonly, plausibly" (*ibid.*, 236) inferred from personal common ground or other inferrable common ground. In other words, the content of inferrable common ground (in Prince's technical sense of "inferrable") is not based in interlocutors' joint attention, unlike personal common ground. Rather, inferrable common ground is only indirectly based in joint attention as a set of plausible inferences which may be drawn from relevant, personal common ground (e.g., hearing a speaker's accent activates a whole host of inferences about their background knowledge; e.g., Weatherhead et al., 2016). To keep this distinction in view while avoiding the false implication that only one type of common is inferred (i.e., Prince's inferrable common ground) and the other not (i.e., personal common ground), following Tomasello (2019) we use the term CULTURAL COMMON GROUND to refer to Prince's technical notion of inferrables. So, both personal and cultural common

ground are inferred from linguistic and nonlinguistic cues. However, they are distinguished by their direct (personal) or indirect (cultural) basis in joint attention. Our decision to use the term cultural common ground reflects Prince's (1981) claim that “the types of inferences needed for the Inferrables [sic] are mostly culture based,” (p. 242) and follows a related naming precedent that distinguishes communal from personal common ground in psycholinguistics (H. H. Clark, 1996).

Does cultural common ground influence children’s referential choice? Presently, no research speaks directly to that question. However, a suggestive study of three-year-olds’ referential interpretation was conducted by Liebal et al. (2013). In that study, an experimenter ambiguously requested one of two dolls – either a Santa Claus doll or an idiosyncratic doll – while appearing either to recognize or not to recognize their intended doll. When the experimenter recognized their intended doll, participants handed her the Santa Claus doll and, when the experimenter did not, they handed her the idiosyncratic doll. One explanation for this pattern is that children formed a plausible link between the experimenter’s facial expression and the differing conventionality of the candidate referents, namely, something like “speakers are more likely to recognize conventional (“old”) than idiosyncratic (“new”) referents, all else equal.” This plausible link may have been the cultural common ground that children used to disambiguate the speaker’s intended referent. If this explanation is correct, then the same kind of plausible link ought to influence children’s referential informativeness, although perhaps at a slightly older age than it influences their comprehension.

To investigate this, we examined four-year-olds’ referential informativeness when playing a novel game with an experimenter (E2) after having learned it from a different experimenter (E1). Children learned with E1 how to use a ball and a stick to operate a marble run. E1 discussed the game either as if it were a conventional activity or an idiosyncratic activity. *Conventional*

1 *discourse* included normative modal verbs (Köymen et al., 2014), a novel name (Rakoczy et al.,
 2 2008), and generic sentences (Roberts et al., 2017) with generic referring expressions (Orvell et
 3 al., 2023; relatedly, Cimpian & Scott, 2012), e.g., “To play Daxing, you don’t use big sticks.” Note
 4 that, while some have argued for it (Gelman & Roberts, 2017; Tomasello, 2019), it is unknown
 5 whether children interpret generic language as communicating cultural common ground (i.e.,
 6 common ground based on cultural group membership). Otherwise, E1 used *idiosyncratic*
 7 *discourse*, in which she invented the game impromptu using specific language, no names, and
 8 verbs of desire or volition, e.g., “To play with this, I don’t want to use this big stick.” After playing,
 9 E1 put the items away. E1 left and a novel adult, E2, entered. To play with E2, children had to
 10 unambiguously refer to the tools. Importantly, E1 had spaced the items closely so that, e.g., the
 11 small stick was only a few inches away from the big stick (an unused distractor). Moreover, the
 12 items were sufficiently out of children’s reach so that the intended referent of a pointing gesture
 13 was ambiguous. This overall configuration meant that, in the absence of common ground,
 14 unambiguous reference required a modified lexical noun (e.g., *the big stick*). Otherwise, with more
 15 common ground, a less informative form may suffice (e.g., *it*). Indeed, two-year-olds’
 16 informativeness reflects this sensitivity to array configuration (O’Neill & Topolovec, 2001). E1
 17 never used modified lexical nouns to refer to the target game items, thus precluding imitation.

18 Children should more readily reason that E2 knows the game items when they
 19 conceptualize the game as conventional compared to idiosyncratic (i.e., because only conventions
 20 are widely known). Indeed, three-year-olds understand conventions as agreements among
 21 knowledgeable agents (Schmidt et al., 2016) and so they, but not two-year-olds, normatively
 22 protest convention violations in novel games (Rakoczy et al., 2008). In fact, three-year-olds restrict
 23 their admonishment to ingroup members (Schmidt et al., 2012), arguably because ingroup

members should know better. Taken with the inverse relation, this suggests that children should be less informative after conventional discourse than after idiosyncratic discourse. Specifically, children may leverage a plausible link like that in Liebal et al. (2013), namely, that listeners are more likely to recognize items in conventional than in idiosyncratic games, all else equal.

Secondarily, we also investigated children's referential choice after E2 used either "we"-framing or "you"-framing when talking to participants. "We"-framing is the selective use of first-person plural pronouns (e.g., "We should play our game!"), whereas "you"-framing substitutes second-person singular pronouns. "We"-framing increases two- and three-year-olds' sense of commitment to partners (Vasil et al., 2024; Vasil & Tomasello, 2022), perhaps by inducing a conceptualization of their partnership from the perspective of "we" (Bacharach, 2006). Based on research that suggests that two-year-olds sometimes overestimate others' knowledge during collaboration (Moll et al., 2011), we thought it possible that "we"-framing may cause children to overestimate their common ground about the novel game with E2. This hypothesis predicts that children should be less informative following "we"-framing compared to "you"-framing. Alternatively, "we"-framing may stoke children's cooperative motivations (Vasil, 2022) by tapping into their predisposition to collaborate (Rekers et al., 2011). This hypothesis predicts that children should be more informative following "we"-framing compared to "you"-framing.

Methods

Participants. There were 124 participants. Twenty-eight participants were excluded. Thus, there were 96 final sample participants ($N = 40$ female, $M = 4;4$, $SD = 0;3$, from 4;0 to 4;11). Exclusion data, race, and income are in Supplementary Tables 1 to 4. Participants were tested in three locations (Supplementary Table 2) by five E1s (Supplementary Table 5). Participants were

recruited in two ways in a moderately sized, southeastern US city. Caregivers with four-year-olds were randomly emailed from a university database of local caregivers who had indicated interest in being contacted about participation in child development research. These children were tested at the university laboratory. Otherwise, in a museum setting, caregivers were asked if they and their child would like to participate. Participation occurred after caregiver written, informed consent and participant verbal assent (protocol 2019-0396 approved by the *X* Institutional Review Board).

Design. The design was 2×2 between-subjects with one repeated measure (24 participants per condition $\times$ 2 observations per participant = 48 observations per condition). One factor was prior discourse by E1 (*conventional discourse, idiosyncratic discourse*). The other factor was linguistic framing by E2 (*“we”-framing, “you”-framing*). Condition assignment was random.

Materials. There were two marbles (one red, one white); a plastic cabinet with 3 drawers (one red, one white, one blue); a handheld marble run; two wooden, red sticks (one short, several inches in length; one long - about 2.5 times the length of the short one); and an easel (Figure 1).



Figure 1. The university laboratory testing area at the start of the familiarization phase. Cabinet on left; easel at center, top; and sticks, marble run, and marbles on the floor on the right.

Procedure. Testing took place in three locations, a laboratory and two locations within a local children's museum (either the museum laboratory or the demonstration space). The museum laboratory was similar to the university laboratory. The museum demonstration space was reserved for exhibits or researchers. However, rather than being an isolated room, the demonstration space was partitioned from the rest of the museum by a short wall over which museum visitors could view the procedure as it was ongoing. In the museum demonstration space, the testing area was isolated from the museum exhibits and visitors.

1 **Warmup Phase.** The warmup occurred outside the testing area. In the university
 2 laboratory, this was a different room from the testing room in which the participant conversed and
 3 played freely with E1 until they appeared comfortable (e.g., responding to, making eye contact
 4 with, or smiling at E1). In the museum, E1 conversed with the participant until they appeared
 5 comfortable. Then, E1 asked the participant if they would like to play “something fun.”
 6 Subsequently, E1 led the participant into the testing area. In the university laboratory and the
 7 museum laboratory, testing took place in a testing room (Figure 1). In both museum testing spaces,
 8 a caregiver was typically present during the procedure but was asked not to interact with the
 9 participant until the procedure ended. In the university laboratory, caregivers were absent during
 10 the remainder of the procedure.

11 **Familiarization Phase.** E1 entered the testing area with the participant. On the ground,
 12 there was a small, colorful marble run with two marbles on one side (one red, one white) and two
 13 red, wooden sticks on the other side (one short, one long; Figure 1). In the conventional condition,
 14 E1 said “Look, there’s Daxing! I’m going to teach you Daxing. Daxing is going to be so much
 15 fun.” For readability, we only describe the conventional condition; the script of the idiosyncratic
 16 condition is paired with the corresponding conventional script in Table 1. E1 and the participant
 17 approached the game. At the game, E1 said “To play Daxing, you don’t use big sticks [*picks up*
 18 *big stick, shows participant*] to push white balls [*picks up white marble, shows participant*].” E1
 19 then got up and put the tools away. While doing this, E1 said “White balls have to go into here
 20 [*places white marble into drawer*] and big sticks over here [*places long stick onto easel*].” E1 put
 21 the white marble into the white drawer of an opaque cabinet with three vertically arranged drawers
 22 (top red, middle white, bottom blue). E1 put the long stick, upright, on the easel.

CULTURAL COMMON GROUND AND REFERENTIAL INFORMATIVENESS

<u>Familiarization phase</u>	
<i>[E1: use unscripted condition-specific discourse as appropriate – see supplementary material for more information]</i>	
<i>In testing room:</i> Idiosyncratic Condition E1: “Look, there’s some toys!” <i>[looks at toys on ground]</i> “I think I’m going to play with these. This is going to be so much fun.”	<i>In testing room:</i> Conventional Condition E1: “Look, there’s Daxing!” <i>[looks at toys on ground]</i> “I’m going to teach you Daxing. Daxing is going to be so much fun.”
<i>E1 and C walk over to game.</i>	
Idiosyncratic Condition E1: “To play with these, I <i>don’t</i> want to use this big stick <i>[picks up big stick]</i> to push this white ball <i>[picks up white ball]</i>.”	Conventional Condition E1: “To play Daxing, you <i>don’t</i> use big sticks <i>[picks up big stick]</i> to push white balls <i>[picks up white ball]</i>.”
<i>E1 goes to put away items.</i>	
Idiosyncratic Condition E1: “I want to put this white ball into here <i>[places white ball into white drawer]</i> and this big stick over here <i>[places big stick onto easel]</i>.”	Conventional Condition E1: “White balls have to go into here <i>[places white ball into white drawer]</i> and big sticks over here <i>[places big stick onto easel]</i>.”
<i>E1 returns to game.</i>	
Idiosyncratic Condition E1: “To play with these, I want to use this <i>[holds up small stick]</i> to push this <i>[holds up</i>	Conventional Condition E1: “To play Daxing, you have to use these <i>[holds up small stick]</i> to push these <i>[holds up</i>

CULTURAL COMMON GROUND AND REFERENTIAL INFORMATIVENESS

<i>pink ball</i>] so it will go down the slide [<i>points to slide</i>]. Look, here's how I want to play [<i>demonstrates once</i>]. See how cool that is?"	<i>pink ball</i>] so they go down the slide [<i>points to slide</i>]. Look, here's how you play Daxing [<i>demonstrates once</i>]. See how cool Daxing is?"
E1: "Here, give it a try!" [<i>hands C ball and stick, C and E play several times</i>]	
E1: "Uh oh, looks like it's time for me to go! Let me see if I can find someone else to play with."	
Idiosyncratic Condition E1: "I'll clean up these , first. This goes into here [<i>places pink ball into pink drawer</i>] and this goes over here [<i>places small stick on the shelf</i>]."	Conventional Condition E1: "I'll clean up Daxing , first. These go into here [<i>places pink ball into pink drawer</i>] and these go over here [<i>places small stick on the shelf</i>]."
E1: "Ok, time for me to go!" [<i>cue for E2 to knock/enter</i>]	
<u>Interim phase</u>	
E2: "Hi! Oh, look!" [<i>points to game</i>]	
E1: "Hi! I was just leaving, but I think [<i>participant's name</i>] wants to keep playing."	
E2: "Ok, bye!"	
E1: "Bye!" <i>E1 leaves</i>	

Table 1. Complete script for E1 in both conditions during the familiarization phase of the experiment. Conditions differences in the E1's language are bolded.

E1 returned to the participant. Now, only the marble run, the red marble, and the short stick were present. E1 said "To play Daxing, you have to use these [*holds up short stick*] to push these

1 [*holds up red marble*] so they go down the slide [*points to slide*]. Look, here's how you play
 2 Daxing [*demonstrates once*]. See how cool Daxing is?" To play, one had to place the marble atop
 3 the run and tap it with the short stick so that it went down the run. One run took about 4 seconds.
 4 After demonstrating once, E1 told the participant "Here, give it a try! [*hands over red marble and*
 5 *short stick*]." The participant took turns playing the game for several minutes.

6 While playing, E1 used condition-dependent or -independent language, contingent on
 7 context. For instance, if the participant suggested alternative play strategies, E1 said "Daxing is
 8 played like this" or "I want to play like this," depending on the discourse condition. In contrast, in
 9 both conditions E1 said "Give it another try!" if the participant hit the marble off the marble run.

10 To conclude playing, in both conditions E1 said "Uh oh, looks like it's time for me to go!
 11 Let me see if I can find someone else to play with. I'll clean Daxing up, first. These go into here
 12 [*places red marble into red drawer*] and these go over here [*places short stick onto easel*]." Now,
 13 only the marble run remained available in the testing area (Supplementary Figure 1). After cleaning
 14 up the red marble and short stick, E1 said "Okay, time for me to go!" Then, E2 knocked on the
 15 door before entering the room. After entering, E2 looked at the participant and E1 and said "Hi!
 16 Oh look!" while excitedly looking at and pointing to the marble run. E1 responded "Hi! I was just
 17 leaving, but I think [*participant name*] wants to keep playing." E2 said "Okay, bye!" Then, E1 left.

18 **Test Phase.** E2 looked excitedly at the participant. In the "we"-framing condition, E2 said
 19 "Do we want to keep playing? What do we think? Do we want to play? We'd have a lot of fun if
 20 we kept playing." The "you"-framing script replaced first-person plural pronouns with second-
 21 person singular pronouns. For example, the "you"-framing script was "Do you want to keep
 22 playing? What do you think? Do you want to play? You'd have a lot of fun if you kept playing."

Participants typically responded immediately and affirmatively to E2's question. However, some participants did not. In such cases, E2 used condition-specific language to pull for an affirmative response, e.g., "Oh, no? But we'd have so much fun if we played!" After responding affirmatively to E2, E2 said "Ok, we should go play, then! Let's [or *You*] go play! We'll have so much fun." E2 walked over to the toy and, while looking between the toy and participant, said "We're going to have a lot of fun playing. Are we ready to play now?"

Recall that only the marble run was available during this exchange. Thus, E2's question pulled for the participant to refer to the marbles and sticks. These items were not immediately available to the participant because they were out of their (but not E2's) reach. The dependent measure was the participant's initial referring forms for the marbles and sticks.

E2 chose a marble or a stick only after the participant unambiguously referred to a marble or a stick, respectively. This required that the participant use the correct modification to distinguish their intended referent (i.e., size for sticks, color for marbles). E2 always chose the items referenced by the participant. Two participants referred to the white marble and 0 referred to the big stick.

If the participant referred immediately to a stick (ambiguously or unambiguously), then E2 walked over to the sticks. Likewise, if the participant referred immediately to a marble, then E2 walked over the marbles. If the participant immediately referred to both items in the same utterance, then E2 walked over to the item referenced first. If the participant did not refer immediately, then after several seconds E2 said (i) "Well, can we play now?" If the participant did not refer after (i), then E2 (ii) walked over randomly to the cabinet or easel and said "Hm..." while looking quizzically at the back half of the side of the cabinet or easel while apparently brushing something off of it with his hand. It was intended that (i) and (ii) would be, arguably, relatively neutral and pragmatically natural means to pull participants to produce referring expressions (i.e.,

“neutral” meaning relatively unsuggestive about E2’s knowledge of the marble run game). If E2 arrived at the cabinet to get the marble, or at the easel to get the stick, and participants still had not unambiguously identified the item, then E2 performed (ii) until participants unambiguously referred to the item. If the participant did not unambiguously refer after several seconds, then E2 delivered the utterance (i), again. If the participant still did not unambiguously refer, (i) was repeated until they did. Four participants never referred to the sticks (then, E2 grabbed the small stick that E1 used). All participants referred to the marbles.

Holding the marble and stick, E2 returned to the marble run. The participant and E2 played the game for several minutes before the procedure concluded.

Coding. There were two observations (“trials”) per participant. Thus, there were 192 trials in the final sample, 188 of which included reference, from 96 participants (i.e., four participants never referred to a stick). For further details of the distribution of references to the game items, please see the Supplementary Material. Participants’ initial referring expressions were coded. Referring expressions were coded for intended referent (“ball” or “stick”); and referring expression type (“no reference,” “no linguistic reference,” “proform,” “unmodified NP,” “modified NP”). In the latter, “no reference” occurred when participants did not refer (these were the four, excluded observations); “no linguistic reference” occurred when participants referred without linguistic form, by pointing; “proform” occurred when participants used personal pronouns (e.g., *it*), null forms (e.g., *over there*), demonstratives (e.g., *that*) or deictic *one* without prenominal modification (but, potentially with demonstratives, e.g., *that one*, or with postnominal modification, e.g., *the one over there* or *one of those*); “unmodified NP” occurred when participants used lexical nouns that lacked prenominal modification (e.g., *the ball*, *a ball*, *that ball*); and “modified NP” occurred when participants used lexical nouns or deictic *one* with prenominal modification (e.g., *the red*

1 *ball* or *the small one*). Reference to the cabinet drawers was coded as marble reference and
 2 categorized following the above (e.g., *the red door* was a modified NP). Postnominal modification,
 3 like *the marble in the red box*, was coded as an unmodified NP.

4 The first author coded the data. A naïve coder coded a random 25% of trials (N = 48 trials).
 5 There was perfect interrater reliability for intended referent (ball or stick), $\kappa = 1.00$, agreement =
 6 100%. There was substantial interrater reliability for referring expression type, agreement =
 7 77.08%, square-weighted $\kappa = .73$. In cases of disagreement, the first author's code was used.

8 **Analysis Plan.** Bayesian regression was conducted in R via brms (Bürkner, 2017). We fit
 9 a hierarchical ordinal model to referring form predicted by framing condition, discourse condition,
 10 their interaction, several covariates, and random effects. One set of analyses compared the
 11 expected informativeness between conditions. A second set of analyses investigated the posterior
 12 probabilities of referring forms. Both sets of analyses used posterior distributions averaged across
 13 all other covariate levels or bins. Priors were weakly informative and fixed effects priors were
 14 centered on zero. For detail about the model, inference, priors, model sampler characteristics,
 15 convergence diagnostics and posterior predictive checks (all adequate per Gelman et al., 2013),
 16 and model posterior parameters, see the Supplementary Material. Code and data to reproduce all
 17 analyses and all data and results Figures are available at the first author's OSF (anonymized link:
 18 https://osf.io/5kqs4/?view_only=73e5ba87e6f44ace955f36b91cfc490f)

20 Results

21 Did conventional discourse or “we”-framing influence participants' linguistic referential
 22 informativeness? If so, which linguistic forms, specifically, were participants more likely to use?
 23 To answer the first question, an ordinal model was fit to the data in Figure 2. In the posterior

model, there was a .96 probability that participants were more informative after conventional discourse compared to idiosyncratic discourse (Figure 3; contrast based on posterior predicted informativeness, described in the Supplementary Methods and visualized in the Supplementary Figure 2). Interestingly, this directional effect was pronounced following “we”-framing, where it held with probability .98, compared to “you”-framing, where it was only .81 (posterior conditional predicted informativeness is displayed in Supplementary Figure 3). In contrast, there was less certainty about the direction of framing effects overall (.86) and following conventional discourse (.90) and idiosyncratic discourse (.65; relevant distributions in Supplementary Figures 4 and 5). Altogether, the direction of discourse effects was counter to our prediction via the inverse pattern. Participants were more informative, rather than less informative, following conventional discourse compared to idiosyncratic discourse, especially when they also heard “we”-framing. In contrast, the data were relatively uninformative regarding the direction of framing effects.

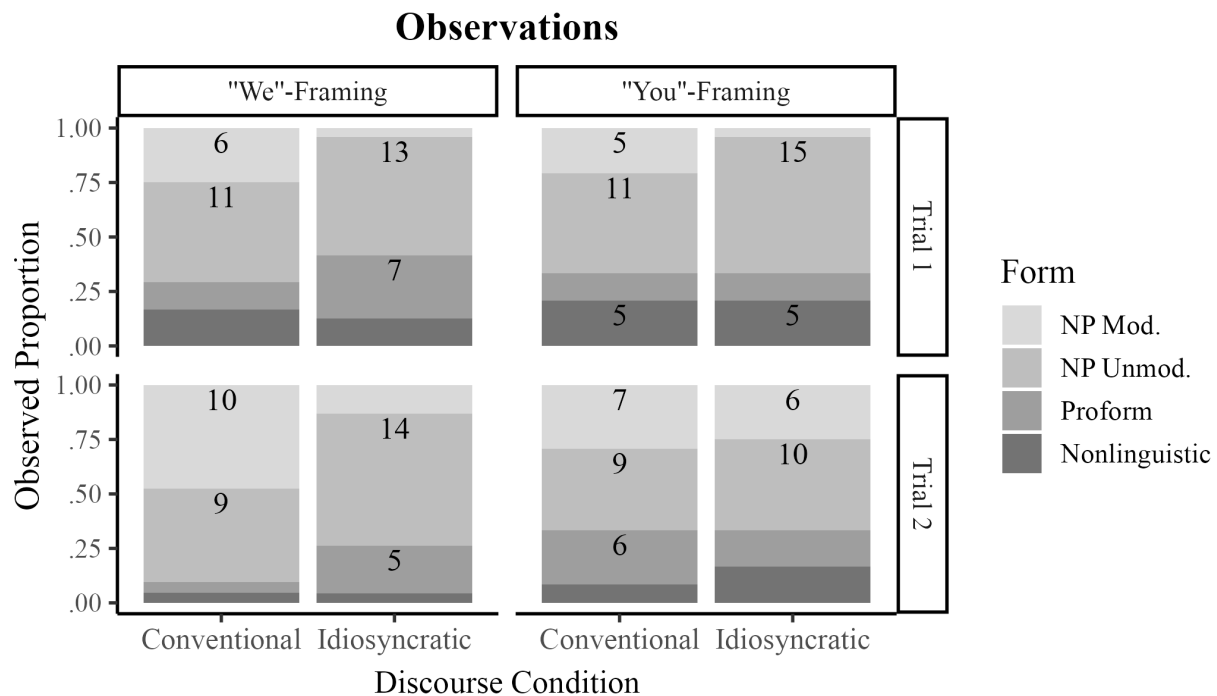


Figure 2. Distribution of 188 observations, excluding four observations of no reference. Inset values represent cell counts. Values less than 5 are hidden.

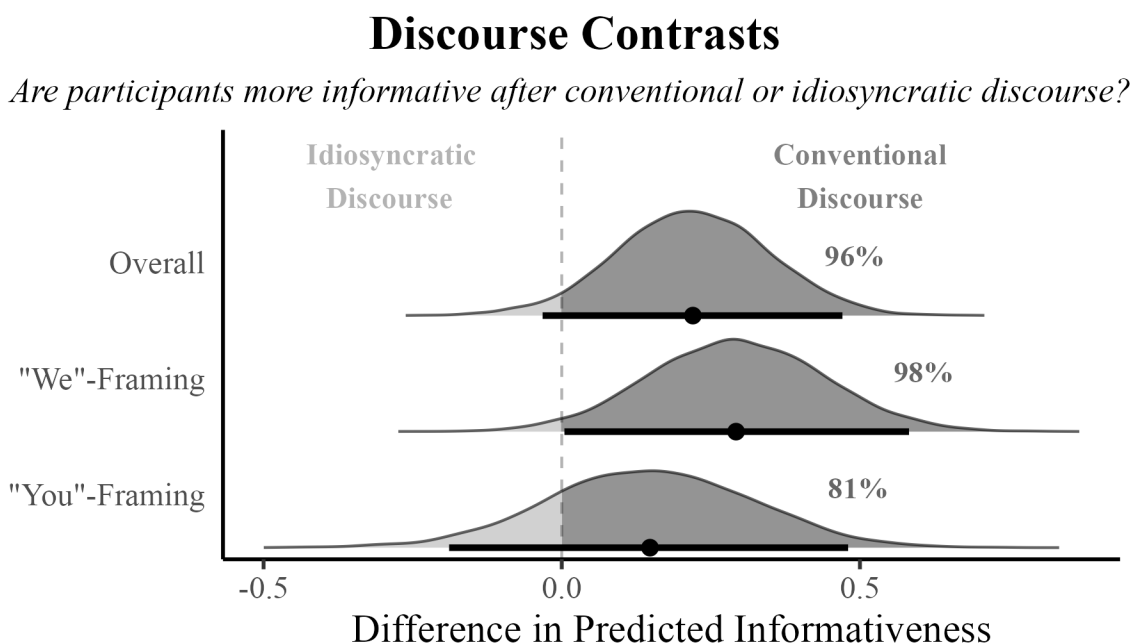


Figure 3. Dots represent means and bars represent 95% highest density intervals. Inset percentages indicate the proportion (multiplied by 100) of posterior draws in which participants were more informative following conventional discourse compared to idiosyncratic discourse. Positive signed density is in dark grey.

Which linguistic forms were participants more likely to use following conventional discourse compared to idiosyncratic discourse? Participants used modified NPs more often following conventional discourse compared to idiosyncratic discourse with probability 1.00 (Figure 4). These same numbers were 1.00 and .98 when paired with “we”-framing and “you”-framing, respectively. This increased use of modified NPs following conventional discourse was offset by a decreased use of unmodified NPs (.03 probability of increased use) and, less reliably,

by decreased use of proforms (.18 overall and .12 and .28 following “we”-framing and “you”-framing, respectively). In contrast, the data were relatively uninformative regarding the direction of the effect of conventional discourse on nonlinguistic reference (probabilities between .34 and .59) and the direction of all framing effects (Supplementary Figure 5). In sum, participants used more modified NPs, and fewer unmodified NPs, following conventional discourse compared to idiosyncratic discourse.

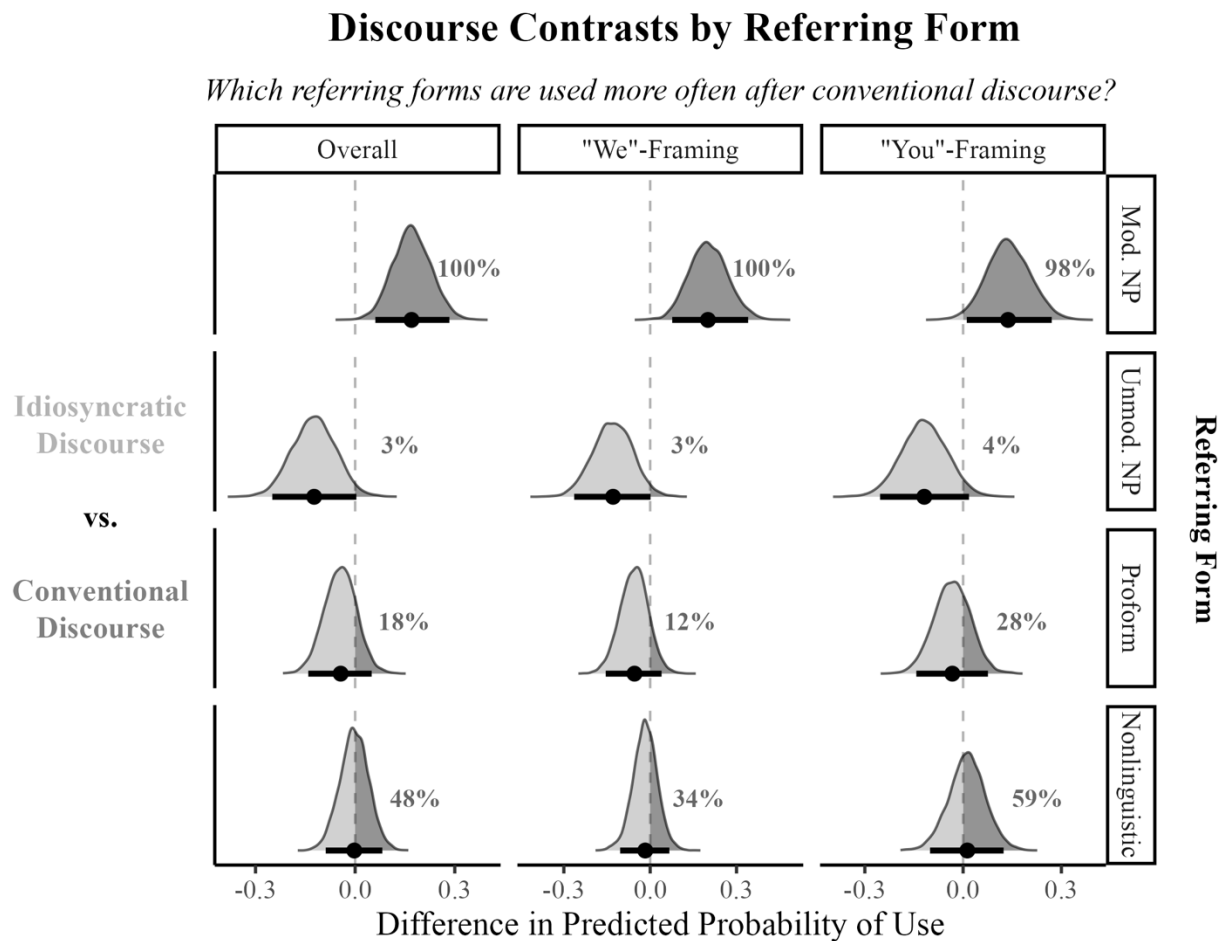


Figure 4. Dots represent means and bars represent 95% highest density intervals. Inset percentages indicate the proportion (multiplied by 100) of posterior draws in which participants were more

likely to produce the referring form following conventional discourse compared to idiosyncratic discourse. Positive signed density is in dark grey.

Taken together, the results for discourse were conclusive, if unexpected. Participants were more informative following conventional discourse compared to idiosyncratic discourse, particularly when it was paired with “we”-framing. Specifically, participants used modified NPs more often, and unmodified NPs less often, after conventional compared to idiosyncratic discourse. The directions of framing effects were poorly informed by the data.

Discussion

This study investigated effects of cultural common ground on children’s referential informativeness. To refer appropriately, children had to infer their common ground (about game items) with a novel individual (E2) based on prior activity with a different, familiar individual (E1). Research on children’s use of personal common ground suggests that children should use less informative referring expressions when they share more cultural common ground with E2, and vice versa. However, contrary to our predictions, we found that children were more informative for E2 when the game they played was introduced by E1 as a conventional activity than when it was a game that E1 ostensibly invented on-the-spot. Specifically, children more often used modified NPs when referring for E2 to items used in the conventional game compared to idiosyncratic game. Moreover, children more often used unmodified NPs for E2 in the idiosyncratic condition compared to the conventional condition. There were no reliable directional effect of linguistic “we”-framing on children’s referential choice. However, children were most informative for E2 when E2 framed their conventional activity with first-person plural pronouns.

Two explanations for the effect of E1's discourse on children's use of NPs focus on the respective roles of motivation and cognition in children's reference (Shatz, 1977). A motivational explanation suggests that conventional discourse enhanced children's motivation to be informative for E2 relative to the idiosyncratic condition. On this explanation, collaboration stokes children's cooperative motives, and conventional games are culturally prepackaged collaborative activities. Consequently, participants who played an ostensibly conventional game were especially cooperatively motivated, and thus more eager to inform their partner. Arguably, the plausibility of this explanation is reinforced by the fact that informativeness was greatest following conventional discourse paired with "we"-framing. This suggests that culturally prepackaged collaboration, plus linguistically framed collaboration, especially stoked children's cooperative motives. Consequently, children in that condition were especially eager to inform their partner in ways leading to success.

On the other hand, a cognitive explanation suggests that it may be (or may also be) that children's informativeness for E2 was dampened by E1's idiosyncratic discourse. In the idiosyncratic condition E1 was, arguably, unnaturally specific about the (red) ball and (big) stick that they wanted to play with. However, someone in a real-world situation may say something more general, like "I want to push the balls down with the sticks." If children themselves found E1 to have been oddly specific, they may have discounted E1's specificity as some idiosyncrasy of E1. Consequently, children may have felt less inclined to use modified NPs to refer to those specific game items for E2. Instead, as the hypothetical real-world example suggests, this explanation predicts that children should use unmodified NPs more often in the idiosyncratic condition (which, indeed, was indeed the pattern that was found). The reason why this explanation predicts, specifically, increased use of unmodified NPs is because the items used in the game

would not be cognitively given (Gundel et al., 1993) or accessible (Ariel, 1988) referents for E2 at the time of children's referential production. Consequently, children should not (and did not) more often use pronouns in the idiosyncratic condition but, rather, unmodified NPs.

Both the motivational and cognitive explanations are plausible and are not mutually exclusive, and future research might look to investigate them. They are preferable to explanations based solely on information structural considerations of referential choice (Lambrecht, 1994), like givenness or accessibility. Such an explanation incorrectly predicts that informativeness will be greater following idiosyncratic discourse compared to conventional discourse, which is the opposite pattern relative to what we found. Likewise, recent Bayesian accounts of referential redundancy (Degen et al., 2020) do not explain our results. Participants rarely produced redundantly modified NPs (also, Sonnenschein, 1985) and it is unclear why redundancy would increase given more common ground.

One limitation of this study was the small amount of linguistic framing by E2 relative to prior discourse by E1. This may have dampened the effect of framing on informativeness for E2. For instance, E1 might have also used “we”-framing. While this would change the referent of “we” relative to the current study (i.e., from a dyad to a group), it might lend children a sense of control or “voice” (Grocke et al., 2018) over the creation of the game and, perhaps, a desire to teach it to ignorant others (Kachel et al., 2018; Köymen et al., 2015; e.g., with more descriptive expressions for game items). Additionally, five people were E1 in three locations. While this is not problematic, in principle, relatively few observations (192) means that this variety may have had an outsize effect on data noisiness. A third limitation is that the behavior of E2 was stilted. This is suggested by the fact that participants were more informative on the second trial than on the first (see the trial effect size visualized in Supplementary Material Figure 6). Furthermore, there was only

CULTURAL COMMON GROUND AND REFERENTIAL INFORMATIVENESS

1 moderately high interrater reliability for referring expressions. This owed to several factors,
2 including pronunciation challenges and the characteristics of our audio recording tools (a GoPro
3 camera's microphone) and testing locations (e.g., ambient noise at the museum demonstration
4 space). Arguably, another limitation was the high level of linguistic skill required for unambiguous
5 reference (modified NPs). This requirement was initially deemed useful as a means to increase the
6 number of ordinal categories.

7 In conclusion, this was the first investigation of the effects of cultural common ground and
8 linguistic “we”-framing on children's referential informativeness. Surprisingly, cultural common
9 ground increased children's informativeness, particularly after linguistic “we”-framing. Arguably,
10 this was because participation in a conventionally organized collaborative activity stoked
11 participants' cooperative motives. There was limited evidence for directional effects of linguistic
12 “we”-framing. Future work may aim to replicate and explore this unexpected result.

References

- Ariel, M. (1988). Referring and Accessibility. *Journal of Linguistics*, 24(1), 65–87.
- Bacharach, M. (2006). *Beyond Individual Choice: Teams and Frames in Game Theory*. Princeton University Press.
- Bohn, M., & Köymen, B. (2018). Common Ground and Development. *Child Development Perspectives*, 12(2), 104–108.
- Bruner, J. S. (1983). *Child's Talk: Learning to Use Language*. Norton.
- Bürkner, P.-C. (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software*, 80(1), Article 1. <https://doi.org/10.18637/jss.v080.i01>
- Cimpian, A., & Scott, R. M. (2012). Children expect generic knowledge to be widely shared. *Cognition*, 123(3), 419–433. <https://doi.org/10.1016/j.cognition.2012.02.003>
- Clark, E. V. (2015). Common Ground. In B. MacWhinney & W. O'Grady (Eds.), *The Handbook of Language Emergence* (First Edition, pp. 328–353). John Wiley and Sons.
- Clark, H. H. (1996). *Using Language*. Cambridge University Press.
- Clark, H. H., & Marshall, C. R. (1981). Definite Reference and Mutual Knowledge. In A. K. Joshi, B. L. Webber, & I. A. Sag (Eds.), *Elements of Discourse Understanding* (pp. 10–63). Cambridge University Press.
- Clark, H. H., & Wilkes-Gibbs, D. (1986). Referring as a collaborative process. *Cognition*, 22(1), 1–39.
- Degen, J., Hawkins, R. D., Graf, C., Kreiss, E., & Goodman, N. D. (2020). When redundancy is useful: A Bayesian approach to “overinformative” referring expressions. *Psychological Review*, 127(4), 591–621. (2020-22960-001). <https://doi.org/10.1037/rev0000186>
- Gauthier, D. (1975). Coordination. *Dialogue*, 14(2), 195–221.

- 1 Gelman, A., Carlin, J., Stern, H., Dunson, D., Vehtari, A., & Rubin, D. (2013). *Bayesian Data*
2 *Analysis* (3rd ed.). CRC Press/Taylor & Francis.
- 3 Gelman, S. A., & Roberts, S. O. (2017). How language shapes the cultural inheritance of
4 categories. *PNAS Proceedings of the National Academy of Sciences of the United States*
5 *of America*, 114(30), 7900–7907. <https://doi.org/10.1073/pnas.1621073114>
- 6 Gilbert, M. (2007). Mutual recognition, common knowledge, and joint attention. *Hommage à*
7 *Wlodek: Philosophical Papers Dedicated to Wlodek Rabinowicz*, 1–21.
- 8 Grice, H. P. (1975). Logic and Conversation. In P. Cole & J. L. Morgan (Eds.), *Syntax and*
9 *Semantics 3: Speech acts* (pp. 41–58). Academic Press.
- 10 Grocke, P., Rossano, F., & Tomasello, M. (2018). Young children are more willing to accept
11 group decisions in which they have had a voice. *Journal of Experimental Child*
12 *Psychology*, 166, 67–78. <https://doi.org/10.1016/j.jecp.2017.08.003>
- 13 Gundel, J. K., Hedberg, N., & Zacharski, R. (1993). Cognitive Status and the Form of Referring
14 Expressions in Discourse. *Language*, 69(2), 274–307. JSTOR.
15 <https://doi.org/10.2307/416535>
- 16 Heim, I. (1983). File Change Semantics and the Familiarity Theory of Definiteness. In R.
17 Bäuerle, C. Schwarze, & A. von Stechow (Eds.), *Meaning, Use, and Interpretation of*
18 *Language* (pp. 164–189). De Gruyter.
- 19 Kachel, U., Svetlova, M., & Tomasello, M. (2018). Three-year-olds’ reactions to a partner’s
20 failure to perform her role in a joint commitment. *Child Development*, 89(5), 1691–1703.
21 <https://doi.org/https://doi.org/10.1111/cdev.12816>
- 22 Karttunen, L. (1968). *What makes definite noun phrases definite?* [RAND corporation report].
23 <https://www.rand.org/content/dam/rand/pubs/papers/2008/P3871.pdf>

- 1 Köymen, B., Lieven, E., Engemann, D. A., Rakoczy, H., Warneken, F., & Tomasello, M. (2014).
2 Children's norm enforcement in their interactions with peers. *Child Development*, 85(3),
3 1108–1122. <https://doi.org/10.1111/cdev.12178>
- 4 Köymen, B., Schmidt, M. F. H., Rost, L., Lieven, E., & Tomasello, M. (2015). Teaching versus
5 enforcing game rules in preschoolers' peer interactions. *Journal of Experimental Child*
6 *Psychology*, 135, 93–101. <https://doi.org/10.1016/j.jecp.2015.02.005>
- 7 Lambrecht, K. (1994). *Information structure and sentence form: Topics, focus, and the mental*
8 *representations of discourse referents*. Cambridge University Press.
- 9 Lee, B. P. H. (2001). Mutual knowledge, background knowledge and shared beliefs: Their roles
10 in establishing common ground. *Journal of Pragmatics*, 33(1), 21–44.
11 [https://doi.org/10.1016/S0378-2166\(99\)00128-9](https://doi.org/10.1016/S0378-2166(99)00128-9)
- 12 Lewis, D. (1969). *Convention: A Philosophical Study*. Blackwell.
- 13 Lewis, D. (1979). Scorekeeping in a language game. *Journal of Philosophical Logic*, 8(1), 339–
14 359. <https://doi.org/10.1007/BF00258436>
- 15 Liebal, K., Carpenter, M., & Tomasello, M. (2013). Young children's understanding of cultural
16 common ground. *The British Journal of Developmental Psychology*, 31(Pt 1), 88–96.
- 17 Moll, H., Carpenter, M., & Tomasello, M. (2011). Social Engagement Leads 2-Year-Olds to
18 Overestimate Others' Knowledge. *Infancy*, 16(3), 248–265.
19 <https://doi.org/10.1111/j.1532-7078.2010.00044.x>
- 20 O'Neill, D. K. (1996). Two-Year-Old Children's Sensitivity to a Parent's Knowledge State When
21 Making Requests. *Child Development*, 67(2), 659–677. [https://doi.org/10.1111/j.1467-](https://doi.org/10.1111/j.1467-8624.1996.tb01758.x)
22 [8624.1996.tb01758.x](https://doi.org/10.1111/j.1467-8624.1996.tb01758.x)

- 1 O'Neill, D. K., & Topolovec, J. C. (2001). Two-year-old children's sensitivity to the referential
2 (in)efficacy of their own gestures. *Journal of Child Language*, 28(1), 1–28.
3 <https://doi.org/10.1017/S0305000900004566>
- 4 Orvell, A., Elli, G., Umscheid, V., Simmons, E., Kross, E., & Gelman, S. A. (2023). Learning the
5 rules of the game: The role of generic “you” and “we” in shaping children's
6 interpretations of norms. *Child Development*, 94(1), 159–171.
7 <https://doi.org/10.1111/cdev.13846>
- 8 Prince, E. (1981). Toward a Taxonomy of Given-New Information. In P. Cole (Ed.), *Radical*
9 *Pragmatics* (pp. 223–255).
- 10 Rakoczy, H., Warneken, F., & Tomasello, M. (2008). The sources of normativity: Young
11 children's awareness of the normative structure of games. *Developmental Psychology*,
12 44(3), 875–881. [https://doi.org/https://doi.org/10.1037/0012-1649.44.3.875](https://doi.org/10.1037/0012-1649.44.3.875)
- 13 Rekers, Y., Haun, D. B. M., & Tomasello, M. (2011). Children, but not chimpanzees, prefer to
14 collaborate. *Current Biology*, 21(20), 1756–1758.
- 15 Roberts, S. O., Ho, A. K., & Gelman, S. A. (2017). Group presence, category labels, and generic
16 statements influence children to treat descriptive group regularities as prescriptive.
17 *Journal of Experimental Child Psychology*, 158, 19–31.
18 <https://doi.org/10.1016/j.jecp.2016.11.013>
- 19 Schiffer, S. R. (1972). *Meaning*. Oxford University Press.
- 20 Schmidt, M. F. H., Rakoczy, H., Mietzsch, T., & Tomasello, M. (2016). Young Children
21 Understand the Role of Agreement in Establishing Arbitrary Norms-But Unanimity Is
22 Key. *Child Development*, 87(2), 612–626. <https://doi.org/10.1111/cdev.12510>

- 1 Schmidt, M. F. H., Rakoczy, H., & Tomasello, M. (2012). Young children enforce social norms
- 2 selectively depending on the violator's group affiliation. *Cognition*, 124(3), 325–333.
- 3 Shatz, M. (1977). The relationship between cognitive processes and the development of
- 4 communication skills. *Nebraska Symposium on Motivation*, 25, 1–42.
- 5 Sonnenschein, S. (1985). The development of referential communication skills: Some situations
- 6 in which speakers give redundant messages. *Journal of Psycholinguistic Research*, 14(5),
- 7 489–508. <https://doi.org/10.1007/BF01666723>
- 8 Stalnaker, R. (1978). Assertion. In P. Cole (Ed.), *Syntax and Semantics* (pp. 315–332). Academic
- 9 Press.
- 10 Stalnaker, R. (2002). Common Ground. *Linguistics and Philosophy*, 25(5–6), 701–721.
- 11 Tomasello, M. (2019). *Becoming Human: A Theory of Ontogeny*. Harvard University Press.
- 12 Vasil, J. (2022). The study of rational framing effects needs developmental psychology. *The*
- 13 *Behavioral and Brain Sciences*, 45, e243. <https://doi.org/10.1017/S0140525X22000930>
- 14 Vasil, J. (2023). A New Look at Young Children's Referential Informativeness. *Perspectives on*
- 15 *Psychological Science*, 18(3), 624–648. <https://doi.org/10.1177/17456916221112072>
- 16 Vasil, J. (2024). *Effects of “We”-Framing and Prior Discourse on Young Children's Referential*
- 17 *Informativeness* [Unpublished PhD Dissertation]. Duke University.
- 18 Vasil, J., Provençal, M., & Tomasello, M. (2024). Effects of “We”-framing and partner number
- 19 on 2- and 3-year-olds' sense of commitment. *Cognitive Development*, 72, 101511.
- 20 <https://doi.org/10.1016/j.cogdev.2024.101511>
- 21 Vasil, J., & Tomasello, M. (2022). Effects of “we”-framing on young children's commitment,
- 22 sharing, and helping. *Journal of Experimental Child Psychology*, 214, 105278.
- 23 <https://doi.org/10.1016/j.jecp.2021.105278>

- 1 Vygotsky, L. S. (1986). *Thought and Language* (4th ed.). The MIT Press.
- 2 Weatherhead, D., White, K. S., & Friedman, O. (2016). Where are you from? Preschoolers infer
- 3 background from accent. *Journal of Experimental Child Psychology*, 143, 171–178.
- 4 <https://doi.org/10.1016/j.jecp.2015.10.011>
- 5